

Anish Aralikatti

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EDUCATION

Indian Institute of Science, Bengaluru

Master of Technology (M.Tech) in Artificial Intelligence

June 2024

GPA: 9.6/10.0

The National Institute of Engineering, Mysuru

Bachelor of Engineering (B.E.) in Computer Science

June 2020

GPA: 9.75/10.0

COURSEWORK DURING MASTER'S DEGREE

Linear Algebra, Probability and Random Processes, Linear and Non Linear Optimization, Data Structures and Algorithms, Machine Learning, Reinforcement Learning, Computer Vision, Generative Models and Representation Learning, Probabilistic and Bayesian Learning, Data Analytics

AWARDS

Secured the highest CGPA in the M.Tech Artificial Intelligent program at IISc, Bengaluru

University Gold Medal for ranking in the top 5 of the Computer Science Department in B.E. at NIE, Mysuru

EXPERIENCE

Cisco Systems, Inc.

Software Engineer

Bengaluru, IN

Aug. 2020 – Present

- Designed and fine-tuned open-source LLMs to create a ChatGPT like system capable of understanding network test frameworks and automating network testing procedures and code generation, enabling users to execute network tests without prior framework knowledge.
- Designed and developed an automated root cause analysis system from multi system network logs using NLP and deep learning to analyse, correlate and identify relevant patterns which reduced the time needed to troubleshoot and fix network problems.
- Developed dynamic data pipelines to extract actionable KPIs from network telemetry data enhancing strategic insights. Built real-time dashboards to improve operational visibility to support stakeholder decision-making. Leveraged machine learning and forecasting to predict network performance, enabling proactive issue resolution and minimizing disruptions.
- Mentored team members providing guidance and support to enhance their skills, improve collaboration and team productivity.

Software Engineer Intern

Jan. 2020 – June 2020

- Worked on network process automation to orchestrate network processes of large enterprise networks.
- Built microservices in NodeJS to automate onboarding and configuration of multiple network devices from a single point. This reduced the configuration time by 95%.
- Developed modules to monitor device health and integrate with controllers such as NSO to take necessary action.

SKILLS

Languages: Python, JavaScript, C++, Java, HTML/CSS, SQL

Frameworks: PyTorch, TensorFlow, LangChain, Django, FastAPI, Angular, NodeJS

Libraries: Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib, NLTK, SpaCy

Tools: Bash Shell, Git, Docker, Kubernetes, AWS, ChromaDB, MongoDB, MilvusDB, Postgres, Redis, Nginx, Kafka, Spark

PROJECTS

Automated Root Cause Analysis from Multi System Logs Using NLP

(NLP, Scikit-Learn, SpaCy, NLTK, Llama, BERT, LLM, VAE, RNN, Transformers)

- Used NLP and deep learning to analyze and correlate logs from various sources, identify relevant patterns and determine the root cause of issues thus reducing the time needed to troubleshoot and fix system problems.
- Utilized models such as RNN and transformers along with LLMs such as Llama and BERT to improve the accuracy and efficiency of log correlation and root cause identification.
- Developed clustering algorithms to group similar logs and classification models to categorize logs based on issue types.
- Used auto correlation analysis using auto associators and semi-supervised learning to perform anomaly detection.
- Implemented real-time log processing capability and scalability of the entire system.

LLM-Driven Network Testing Automation

(LLM, Llama, Robot Framework, Knowledge Graphs, Retrieval Augmented Generation (RAG), NLP)

- Designed and implemented a system that leverages large language models (LLMs) to interpret user queries for automated network testing. The system transforms natural language inputs into Robot Framework test files by dynamically generating scripts to execute network tests without the need for manual coding.

- Utilized knowledge graphs to validate input queries to ensure the accuracy and correctness of each query before test generation. This reduces errors and ensures the generated tests are correct.
- Implemented Retrieval Augmented Generation (RAG) to access relevant documentation and generate code snippets for test cases. These snippets are then used to initiate and monitor the test execution on network devices followed by the automated creation of reports based on test outcomes.
- Incorporated validation checks and interactive prompts for users to provide additional details or corrections, enhancing test accuracy and user experience.
- This system significantly reduces the time and expertise required for network test automation, enabling non-technical users to initiate complex network tests through simple prompts. It increases testing efficiency, improves accuracy, and enhances accessibility in network operations.

Depression Detection using Speech and Text Analysis

(NLP, Scikit-Learn, PyTorch, Tensorflow, SpaCy, NLTK, Matplotlib, Seaborn)

- Developed a deep learning model for detection of depression by analyzing speech patterns and textual data.
- Employed acoustic feature extraction for speech and sentiment analysis for text to capture relevant features that are indicators of depressive symptoms.
- Built and trained models including Support Vector Machines (SVM), Convolutional Neural Networks (CNN), LSTM, BiLSTM and Transformers to classify depression levels based on the extracted features.
- Validated on test data to ensure the accuracy and reliability of the trained depression detection model.

Human Activity and Behaviour Detection in Crowded Scenes

(PyTorch, OpenCV, C3D, I3D, ResNet3D, Multi-task Learning, Transfer Learning)

- Utilized deep learning techniques to detect crowd behavior in public spaces during social protests, cultural festivals, concerts, urban gatherings, etc to improve crowd management strategies.
- Used video camera feed as input along with corresponding optical motion maps and developed a Multi-Input Multi-Task Learning framework for classification of crowd behaviour and crowd motion flow.
- Tried various backbone network models like C3D, I3D, and ResNet3D for video feature extraction and experimented with different numbers of shared layers to assess performance of the model.
- Introduced the Crowd-11 Behavior Dataset, an extension of Crowd-11 dataset, providing behavior labels alongside existing motion labels for each video clip to help in behaviour classification.

Crowd Segmentation and Counting using Few Shot Learning

(PyTorch, OpenCV, Computer Vision, Few-Shot Learning, Meta-learning)

- Used a Few-Shot Learning approach to tackle the fundamental problem of crowd counting in densely crowded scenes.
- Utilized meta-learning in the context of few-shot learning to enhance model adaptation to target crowd scenes with only a few training samples. Thus overcame the challenges related to the limited availability of annotated data and the constraints of conventional supervised learning methods that depend on the requirement of vast amounts of labeled training data.
- Trained the model for rapid adaptation to target crowd scenes with minimal examples during testing, ensuring it generalizes well to new scenes and performs well on the task of crowd counting.

Community Detection and Link Prediction in Social Networks

(Python, Scikit-learn, NetworkX, NumPy, XGBoost, Graph Neural Networks (GNN))

- Performed community detection on a Facebook dataset by implementing the Louvain algorithm from scratch and applied spectral clustering to analyse the structural properties and relationships within the detected communities.
- Implemented graph machine learning techniques like GNNs for predicting missing links in the network.
- Used the Metropolis-Hastings sampling technique to sample positive and negative samples to train the model effectively.
- Incorporated graph measure features to enrich the feature set for link prediction and was able to get a ROC-AUC score of 0.96 for link prediction between nodes.

Real-time Objection Detection and Face Recognition System to assist the Visually Impaired

(Python, Java, OpenCV, YOLOv3, FaceNet, TensorFlow, Android)

- Developed an Android app that enables real-time object detection and face recognition on smartphones to assist visually impaired users in identifying nearby objects and people.
- Utilized OpenCV for processing live video feeds, implemented object detection with YOLOv3 and integrated FaceNet for face recognition when faces are identified. Trained the system using TensorFlow on a dataset of known individuals.
- Provided audio feedback via Google's text-to-speech API, prioritizing important objects using a priority queue.

PUBLICATIONS

- Aralikatti, Anish, et al. "Real-time object detection and face recognition system to assist the visually impaired." *Journal of Physics: Conference Series*, Vol. 1706, No. 1. IOP Publishing, 2020.

Highlights: Developed a low-latency computer vision pipeline in a resource constrained environment integrating object detection and facial recognition to aid visually impaired individuals in real-time navigation and social interaction.

<https://iopscience.iop.org/article/10.1088/1742-6596/1706/1/012149>

- Latapie, Hugo, Aralikatti, Anish, et al. "Ethosight: A Reasoning-Guided Iterative Learning System for Nuanced Perception based on Joint-Embedding & Contextual Label Affinity." (2023).

Highlights: Co-developed a novel reasoning-guided iterative learning system for nuanced video perception. Trained joint-embedding models and context-aware label-affinity metrics to iteratively refine semantic labels, enabling adaptive, zero-shot recognition of complex behaviors